JIWON JUNG

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RESEARCH INTERESTS

Stochastic modeling, statistical inference, and machine learning for dependent time series and event data; point processes and Hawkes models, with applications in finance, healthcare, and manufacturing.

EDUCATION

Ph.D. in Statistics, Purdue University 2021–2024
Dissertation: Dynamics of Modern Financial Markets: Data-Driven Approaches
Co-Advisors: Prof. Kiseop Lee and Prof. Mengyi Xu

M.S. in Statistics, Seoul National University 2017–2019

B.S. in Statistics and **B.A. in Economics,** Seoul National University 2013–2017

PROFESSIONAL EXPERIENCE

Academic Appointments

University of Washington, Applied Mathematics: Pearson Fellow/Acting Instructor Sep. 2025–Present
Purdue University, Statistics: Visiting Assistant Professor Jan. 2025–Aug. 2025

Research & Industry Experience

Vivity AI: Smart Factory Data Analyst (Project-Based University–Industry Collaboration) 2022–2024
Asan Medical Center: Cancer Center Research Assistant (Seoul, Korea) 2019–2020
LG CNS: Smart Factory Winter Intern (Seoul, Korea) 2018

SELECTED AWARDS & HONORS

I.W. Burr Award for Excellence in Dissertation Research and Results, Purdue University 2025
Outstanding Teaching Award for Teaching Assistant, Purdue University 2024
Recognition Award for Efficiency Improvements, Purdue University 2024
Emily and Paul Kidwell Graduate Student Excellence Award, Purdue University 2022

RESEARCH PAPERS

Working Papers

1. **Jung, J.** and Leung, T. (2026). Modeling the Interactions Between Zero-Day Options and Underlying Markets Using Joint Limit Order Book Data. Manuscript under review; SSRN Abstract No. 7097018.
2. **Jung, J.** and Lee, K. (2026). Common Signals, Spurious Cross-Excitation, and Identification in Multivariate Hawkes Processes. Manuscript under review; arXiv:2610.19428
3. **Jung, J.** and Lee, K. (2026). Do Round Numbers Hold Prices? Occupation Densities and Depth-Conditioned Hazards in the Limit Order Book. Manuscript under review; arXiv:2610.34787

Peer-Reviewed Publications

1. **Jung, J.**, Lee, K., and Xu, M. (2026). Modeling Multistate Health Transitions with a Most-Recent-Event Hawkes Process. *North American Actuarial Journal*, 30 (1), 157–178.
2. **Jung, J.** and Lee, K. (2025). Attention-Based Reading, Highlighting, and Forecasting of the Limit Order Book. *Quantitative Finance*, 25 (7), 1015–1027. [**Feature Article**]
3. **Jung, J.**, Lee, K., and Leung, T. (2025). Threshold Overnight Comovement Analysis of Intraday and Overnight Returns. *Investment Analysts Journal*, 55 (2), 109–129.

4. Ho, D. J., Chui, M. H., Vanderbilt, C. M., **Jung, J.**, Robson, M. E., Park, C. S., and Fuchs, T. J. (2023). Deep Interactive Learning-Based Ovarian Cancer Segmentation of H&E-Stained Whole Slide Images to Study Morphological Patterns of BRCA Mutation. *Journal of Pathology Informatics*, 14, 100160.

5. **Jung, J.**, Ha, S., Son, W., Lee, J., and Won, J. H. (2022). SportLight: Statistically Principled Crowdsourcing Method for Sports Highlight Selection. *Journal of the Korean Statistical Society*, 51 (1), 127-148.

6. Shin, S. J., You, S. C., Jeon, H., **Jung, J.** W., An, M. H., Park, R. W., and Roh, J. (2021). Style Transfer Strategy for Developing a Generalizable Deep Learning Application in Digital Pathology. *Computer Methods and Programs in Biomedicine*, 198, 105815.

7. Kim, S. W., Roh, J., **Jung, J.**, Pak, H. K., Lee, A. N., Park, Y. S., and Park, C. S. (2020). Immune Checkpoint Molecule V-set Ig Domain-Containing 4 (VSIG4) Expression is Associated with Poor Prognosis in Advanced Gastric Cancer Patients. *The Journal of Immunology*, 204, 243.4-243.4

8. Roh, J., **Jung, J.**, Lee, Y., Kim, S. W., Pak, H. K., Lee, A., and Park, C. S. (2020). Risk Stratification Using Multivariable Fractional Polynomials in Diffuse Large B-Cell Lymphoma. *Frontiers in Oncology*, 10, 329.

CONFERENCE PRESENTATIONS

1. **Jung, J.** (2027). From Stock to SoC: Advising Students Across Modeling Domains. *Joint Mathematics Meetings (JMM) 2027*. Accepted for presentation in the Special Session on Mathematical Modeling in Action.

2. **Jung, J.** and Leung, T. (2026). Modeling the Interactions Between Zero-Day Options and Underlying Markets Using Joint Limit Order Book Data. Accepted for presentation, *Job Market Showcase Talk at INFORMS Annual Meeting 2026*, San Francisco, CA, U.S.

3. **Jung, J.** and Lee, K. (2024). Attention-Based Reading, Highlighting, and Forecasting of the Limit Order Book. *Invited talk at Joint Statistical Meetings 2024*, Portland, OR, U.S.

4. **Jung, J.**, Leung, T., and Lee, K. (2024). Threshold Overnight Comovement Analysis of Intraday and Overnight Returns. *Invited talk at SERI 2024*, Sungkyunkwan University, Seoul, Korea

5. **Jung, J.**, Lee, K., and Xu, M. (2024). Modeling Multi-state Health Transitions with a Self Exciting Process. *Invited talk at American Mathematical Society (AMS) Sectional Meeting*, University of Wisconsin-Milwaukee, Milwaukee, WI, U.S.

6. **Jung, J.**, Lee, K. (2023). Attention-Based Reading, Highlighting, and Forecasting of the Limit Order Book. *Invited talk at 2023 SKKU-KAFE conference*, virtual

7. **Jung, J.**, Lee, K., and Xu, M. (2023). Modeling Multi-state Health Transitions with Hawkes Processes. *Invited talk at INFORMS 2023*, Phoenix, AZ, U.S.

8. **Jung, J.**, Lee, K., and Xu, M. (2023). Modeling Multi-state Health Transitions with Hawkes Processes. *CEPAR International Conference*, UNSW, Sydney, Australia

9. **Jung, J.** and Lee, K. (2023). Attention-Based Reading, Highlighting, and Forecasting of the Limit Order Book. *Invited talk at SIAM Financial Mathematics and Engineering 2023*, DoubleTree by Hilton Philadelphia Center City, Philadelphia, PA, U.S.

10. **Jung, J.**, Lee, K., and Xu, M. (2023). Modeling Multi-state Health Transitions with Hawkes Processes. *Invited talk at Ajou University WCU conference*, Ajou University, Suwon, Korea

11. **Jung, J.**, Leung, T., and Lee, K. (2023). A Lead-lag Analysis of Intraday and Overnight Returns. *Invited talk at American Mathematical Society (AMS) Sectional Meeting*, Georgia Institute of Technology, Atlanta, GA, U.S.

12. **Jung, J.**, Lee, K., and Xu, M. (2022). Modeling Functional Disability with Hawkes Process. *Actuarial Research Conference*, Urbana, IL.

13. **Jung, J.**, Roh, J., and Park, C. S. (2021). Abstract PO-079: Fused LASSO application for gastric cancer image segmentation. Clinical Cancer Research, 27, PO-079. *American Association for Cancer Research*, virtual.

TEACHING EXPERIENCE

Instructor

University of Washington, Applied Mathematics

— CFRM 405: Mathematical Methods for Quantitative Finance — Fall 2025; Fall 2026
— CFRM 421/521: Machine Learning for Finance — Spring 2026
— CFRM 425: R Programming for Quantitative Finance — Winter 2026
— CFRM 590: Multiscale Financial Signal Processing (co-instructor: Prof. Tim Leung) — Fall 2025

Purdue University, Statistics

— STAT 517: Statistical Inference (online) — Summer 2025
— STAT 301: Elementary Statistical Methods (10 sections incl. Indianapolis campus) — Spring 2023–2025

Teaching Assistant

Purdue University, Statistics: STAT 303, 511, 512 (2021–2022).
Seoul National University: Statistics, Seminar: Information Theory, Seminar: Knowledge, Science Camp, and Big Data Special Course using R (2016–2018).

PROFESSIONAL SERVICE AND DEVELOPMENT

Professional Service

Ad hoc reviewer: *ASTIN Bulletin; Computational Statistics and Data Analysis; European Journal of Finance; Journal of Data Science; Studies in Economics & Finance* — 2024–2026
Co-organizer, invited session: "Data-Driven Methods in Financial Markets," JSM 2024 — 2024

Mentoring and Student Service

Advised a three-student team for the COMAP Mathematical Contest in Modeling — 2026
Mentored Purdue Undergraduate Research Conference participant — 2025
Served as judge for undergraduate research presentations — 2025

Travel Grants

George Casella Travel Award and ASA Travel Fund for JSM (2024); CEPAR Travel Grant (2023); SIAM Travel Grant (2023); Graduate Women in Science Program, Purdue University (2022); Graduate Student Instructor scholarship, Seoul National University (2018); Academic Scholarship for Excellence (2013–2014, 2018)

TECHNICAL SKILLS & LANGUAGES

Programming languages: `Python`*, `R`*, `MATLAB`, `Julia`, `C/C++`, `Java` *Used in teaching.
Software and tools: SPSS*, Excel*, Bloomberg Terminal*
Languages: English (fluent), Korean (native)